

[PZoom Deep Dive 9.3] Exhaust the contradiction where γ is a proper subset of both

Assume for contradiction that $\gamma \neq \alpha$ and $\gamma \neq \beta$.

Because $\gamma = \alpha \cap \beta$, we know that $\gamma \subseteq \alpha$ and $\gamma \subseteq \beta$. Since we are assuming γ is unequal to both, it must be a strict proper subset of both:

$$\gamma \subsetneq \alpha \quad \text{and} \quad \gamma \subsetneq \beta.$$

By Lemma 1 (Structure of Transitive Subsets), since γ is a transitive proper subset of the ordinal α , it must be an element of α . Similarly, since γ is a transitive proper subset of the ordinal β , it must be an element of β :

$$\gamma \in \alpha \quad \text{and} \quad \gamma \in \beta.$$

By definition of the intersection, an object belonging to both sets must belong to their intersection. Therefore:

$$\gamma \in (\alpha \cap \beta) \implies \gamma \in \gamma.$$

This states that γ is a member of itself. We can now construct the singleton set $K = \{\gamma\}$. Because $\gamma \in \gamma$, the set K has no $\in$ -minimal element: its only member shares an element with K . This directly violates the Axiom of Regularity, which states that every nonempty set must contain an element disjoint from it under membership.

Thus, the assumption that γ is a proper subset of both ordinals is untenable. One of the sets must match γ entirely.

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